

A Quantitative Proteomic Analysis of Light Adaptation in a Globally Significant Marine Cyanobacterium *Prochlorococcus marinus* MED4

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In this study, we conducted biological and technical replicate proteomic experiments using isobaric tags for relative and absolute quantification (iTRAQ), to elucidate the light adaptation strategies of *Prochlorococcus marinus* MED4. The MED4 strain is adapted to an oceanic environment characterized by low nutrient levels, and ever-changing light intensities. Approximately 11% of the proteome was identified, with an average coefficient of variation of iTRAQ quantification values of 0.15. Fifteen proteins were deemed to be statistically and significantly differentially expressed in changing light intensities, particularly the down-regulation of photosystem-related proteins, and the up-regulation of the stress-related chaperone GroEL in high light compared to low light.

Keywords: iTRAQ • LC–MS/MS • high light • *Prochlorococcus* • photoadaptation

Introduction

Cyanobacteria are oxygenic phototrophic bacteria known to thrive in conditions previously thought to be intolerable, resisting desiccation,¹ high temperatures,² extreme pH,³ and hypersalinity.⁴ In 1988, the cyanobacterium *Prochlorococcus* was discovered⁵ and found to reside in nutrient-poor oceans, which were also thought to be void of living microorganisms. Since their discovery in this hostile habitat, the interest in their ecological impact has increased greatly, mostly because they are believed to be the most abundant known photosynthetic organism on earth,¹⁰ and therefore, the substantial contribution of primary production⁶ from *Prochlorococcus* means that they play an important role in nutrient cycling and global climate control.

The increasing interest in *Prochlorococcus* has meant that five different strains have now been fully sequenced (http://genome.jgi-psf.org/mic_home.html and <http://www.genoscope.cns.fr/>), assisting genetic studies^{7–11} into environmental adaptation strategies. Fully sequenced microorganisms also make attractive candidates for proteomic analysis. Furthermore, it is rarely feasible to use the nucleotide sequence of an organism to predict ecological function and environmental adaptation, without information on which proteins are synthesized in specific conditions. Therefore, a quantitative proteomic analysis of the challenges *Prochlorococcus* faces in its environment has the potential to reveal mechanisms of adaptation in vast oceanic niches.

Prochlorococcus marinus is the smallest known photosynthetic organism, with a cell size around 0.5–0.7 μm .⁹ It is characterized by a novel complement of pigments,⁹ including divinyl derivatives of chlorophyll a and b (chl a_2 and b_2). Unlike

all other cyanobacteria that harvest light using phycobilisomes (thylakoid membrane extrinsic multisubunit complexes), its major light harvesting complex is a protein which binds chl a_2 and b_2 , situated in the thylakoid membrane.¹² Remarkably, *Prochlorococcus* can be found on the surface of the ocean where levels of light are high, and up to 200 m below where natural light is in very small quantities. It is able to adapt to these varying light conditions due to different ecotypes that differ in pigment composition.¹³ The low-light ecotype grows optimally where the amount of light can be as low as 1 $\mu\text{mol quanta m}^{-2} \text{ s}^{-1}$, and is characterized by a high ratio of chl b_2/a_2 and the possession of multiple antenna genes (*pcb* genes).¹⁴ The genotype of the most distantly related high-light ecotype is only 2.7% genetically distinct to the low-light ecotype,¹³ but contains a low ratio of chl b_2/a_2 and only one *pcb* gene.¹⁴ High-light ecotypes are dominant at the ocean surface, but can also survive deeper into the ocean where levels of light are much lower. The availability of solar light is therefore a major controlling factor for the ecological distribution of *Prochlorococcus* strains.^{11,15,16}

MED4 is a high-light-adapted strain of *Prochlorococcus*, and its genome is fully sequenced and annotated. At 1.66 Mbp, it is much smaller than the largest sequenced *P. marinus* strain MIT9313 (2.4 Mbp).¹⁰ In relation to its preferred environmental conditions, MED4 possess many more high-light inducible proteins (HLIPs) and photolyases to repair DNA damaged by UV light, compared to its low-light counterparts.¹⁷ MED4 has one of the highest A + T contents (69.2%)⁷ for bacteria, and this means it is more susceptible to DNA damage caused by UV-induced thymine dimers. Whether cells have specific mechanisms to overcome this sensitivity, or whether the increased likelihood of mutations is an advantage to *P. marinus* MED4 strains, remains unclear. Furthermore, the surface ocean waters are subject to more rigorous mixing than deeper waters,

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and this means MED4 cells must adapt to rapidly changing irradiances.¹⁸

Photosynthetic organisms must adapt to varying light intensities for survival. If photosynthesis and the components involved are not correctly regulated, cellular processes can become damaged due to the production of reactive oxygen species caused by the imbalance of energy supply and dispersion.¹⁹ The immediate cellular response to high light includes enhanced repair, changes in energy transfer from the light harvesting complex to photosystem II (PSII), reduced light harvesting antennae, state transitions, and protective energy dissipation.^{20–22} The cellular effects of light intensity have been studied in the cyanobacterium *Synechocystis* sp. PCC 6803 using DNA microarray analysis,²³ as well as the response of several cyanobacteria from the perspective of stress-associated genes.²⁴

A very recent proteomic study was conducted on the alga *Chlamydomonas reinhardtii* investigating the effects of high-light stress, where a two-dimensional electrophoresis (2DE) approach was used for protein separation and visualization and peptide mass fingerprinting was used for identification.²⁵ Interestingly, only two proteins (heat shock protein and oxygen-evolving enhancer protein) out of the 32 identified and differentially expressed proteins were previously associated with high-light stress, albeit through transcriptomic studies. Differentially regulated proteins were carbon fixation related, molecular chaperones, or light harvesting complex proteins.²⁵ Other proteomic studies have examined the high-light response of a specific cellular compartment, for example, the thylakoid proteome in *Arabidopsis*,²⁶ where the role of oxidative stress defense was prominent.

In this paper, we seek to examine the adaptation of *P. marinus* MED4 to varying levels of light intensity, imitating natural oceanic conditions, from a proteomic perspective. This is the first proteomic analysis, to our knowledge, which assesses the adaptation of a marine cyanobacterium to light conditions experienced in the environment. We employ the isobaric tags for relative and absolute quantification (iTRAQ) technique and conduct subsequent bioinformatics analysis using ProQuant and ProGroup software for protein quantification, identification, and relative differential expression. iTRAQ is powerful in that it enables the comparison of several phenotypes in one experiment and also permits the inclusion of technical and biological replicates, and quantification occurs at the MS/MS stage.²⁷ iTRAQ has been demonstrated as a useful, robust, and reliable tool for proteomic studies,²⁷ with several advantages over densitometry and labeling proteomic techniques.^{28,29} Essentially, through this study, we aim to further understand the photoacclimation of a globally abundant marine cyanobacterium.

Material and Methods

All chemicals were purchased from Sigma-Aldrich (Gillingham, Dorset, U.K.) unless otherwise stated.

Culture Conditions and Cell Growth. *P. marinus* MED4 was provided by the Roscoff Culture Collection (RCC) and grown in seawater-based Pro99 medium.³⁰ It was originally isolated from the northwest Mediterranean Sea in 1989 from surface waters and cultured in PCR-S11 medium. The seawater used for the media was obtained locally from the coastline of Hull, U.K. (see 'Seawater Comparison' below). The microorganism was cultured at 24 °C in 250 mL flasks, in 12 h light/dark cycles and under continuous shaking (220 rpm). Three separate cultures differing in light intensity were grown, low (20 μ ein-

stein $\text{m}^{-2} \text{s}^{-1}$), medium (60 μ einstein $\text{m}^{-2} \text{s}^{-1}$), and high (100 μ einstein $\text{m}^{-2} \text{s}^{-1}$). These were subcultured three times in late-exponential phase (ca. 30 days) to ensure adaptation to the different conditions. Light measurements were made using a Loggerhead 2100 light probe (Biospherical Instruments, San Diego, CA). Optical density measurements were used to follow cell growth at the same time of day at a wavelength of 446 nm using an UltraSpec 2100 Pro spectrophotometer (Amersham Biosciences, Buckinghamshire, U.K.). Growth was measured at the same time of day because the cell cycle is tightly linked to the light/dark phase.³¹

Seawater Comparison. Levels of major and minor trace elements in local seawater (Hull, U.K.) and English Channel seawater (Roscoff) were compared using inductively coupled plasma mass spectrometry (ICP-MS, Perkin-Elmer, Elan 400). Levels of 18 trace elements and 5 major elements (Ca, Fe, K, Mg, and Na) were compared. A detailed composition of seawater³² was used to estimate the standard addition ranges of a multielement standard Merck VIII (Merck, Darmstadt, Germany) required for accurate abundance measurements. For both seawater samples, two dilution sets were made, a 500:1 and a 1000:1 dilution. There was either no standard added or 100, 300, 500, 1000, 3000, or 5000 ppb of the standard added. These standard additions were used for measuring major elements. For trace elements, the same dilutions of seawater (500:1 and 1000:1) were used with again no standard added, and with 0.1, 0.3, 0.5, 1.0, 3.0, and 5.0 ppb standard added. A linear relationship between the concentration of element added and the measured signal (cps) was used to calculate the concentration of the elements.

Protein Extraction and iTRAQ. In total, 50 mL of cells was harvested at mid-exponential phase and centrifuged (Heraeus Model Multifuge 3S-R, Germany) at 10 000g for 20 min at 4 °C. To aid protein extraction and remove salts and polysaccharides, cells were washed with 5 vol of a sucrose-based buffer consisting of 50 mM Tris (pH 7.4), 100 mM EDTA, and 25% (w/v) sucrose. For protein extraction, pelleted cells were re-suspended in 1 mL of freshly made extraction buffer consisting of 9 M urea, 4% (w/v) CHAPS, 1% (w/v) DTT, 0.8% (w/v) Pharmalyte, 5% (w/v) protease inhibitor cocktail, and 10 mM PVPP. Proteins were extracted using mechanical cracking and liquid nitrogen,³³ and proteins were recovered by centrifugation at 21 000g for 30 min at 4 °C. The total protein concentration obtained was quantified using the RC DC Protein Assay (Bio-Rad, Hertfordshire, U.K.) according to the manufacturer's instructions. A total of 100 μ g of protein from all three light conditions including a high-light duplicate (technical replicate) was precipitated using TCA/acetone. This was done by mixing the exact volume of protein solution corresponding to 100 μ g of protein with 3 vol of acetone with 5% TCA and 20 mM DTT, and incubating at –20 °C overnight. After incubation, the protein was recovered by centrifugation at 21 000g at 4 °C for 30 min. The resulting pellet was washed with ice-cold acetone and resuspended in 30 μ L of 500 mM TEAB buffer. Each sample was reduced, alkylated, digested, and labeled with iTRAQ reagents according to the manufacturer's (Applied Biosystems) protocol with modifications³⁴ (Table 1). After labeling, samples were combined and dried in a vacuum centrifuge (Model 5301, Eppendorf, Cambridgeshire, U.K.) at room temperature.

Strong-Cation Exchange Fractionation. Peptides were re-suspended in 200 μ L of aqueous buffer A (10 mM KH_2PO_4 and 25% acetonitrile at pH 3.0, filter sterilized). Separation was performed on a BioLC HPLC unit (Dionex, Surrey, U.K.) using

Table 1. Different Phenotypes Labeled with iTRAQ Label Reagents^a

experiment	iTRAQ label reagent			
	114	115	116	117
1	HL	LL	ML	HL
2	LL	ML	HL	HL

^a Technical replicates are shown in bold type. HL, high light at 100 $\mu\text{Einstein m}^{-2} \text{ s}^{-1}$; ML, medium light at 60 $\mu\text{Einstein m}^{-2} \text{ s}^{-1}$; LL, low light at 20 $\mu\text{Einstein m}^{-2} \text{ s}^{-1}$.

a PolySULFOETHYL A column (PolyLC, Columbia, MD), 5 μm of 100 mm length $\times$ 2.1 mm i.d. and 200 $\text{\AA}$ pore size. Buffer B consisted of 10 mM KH_2PO_4 , 25% acetonitrile, and 500 mM KCl at pH 3.0, filter sterilized. The 60 min gradient consisted of 100% buffer A for 5 min, 8–24% buffer B for 40 min, 24–100% buffer B for 5 min, 100% buffer B for 5 min, and finally 100% buffer A for 5 min. A flow rate of 0.2 mL/min was maintained with an injection volume of 200 μL . A UV detector UVD170U and Chromeleon Software v6.50 (Dionex/LC Packings, The Netherlands), was used to monitor the chromatogram as fractions were collected every minute using a Foxy Jr. Fraction Collector (Dionex). Fractions were pooled depending on the peak intensities in the chromatogram and then dried in a vacuum centrifuge. Samples were stored at -20°C prior to mass spectrometric analysis.

Mass Spectrometry and Protein Identification. Mass spectrometry was performed using a QStar XL Hybrid ESI Quadrupole time-of-flight tandem mass spectrometer, ESI-qQ-TOF-MS/MS (Applied Biosystems, Framingham, MA; MDS-Sciex, Concord, Ontario, Canada) coupled with an online capillary liquid chromatography system (Famos, Switchos and Ultimate from Dionex/LC Packings, Amsterdam, The Netherlands). A volume of 55 μL of Switchos buffer (0.1% formic acid and 3% acetonitrile) was used to resuspend dried samples ready for the MS, and 20 μL was injected to the nanoLC-ESI-MS/MS system for each analysis. Initial separation took place on a PepMap C-18 RP capillary column (LC Packings) with a constant flow rate of 0.3 $\mu\text{L}/\text{min}$. LC buffers A and B were 3% acetonitrile and 0.1% formic acid, and 97% acetonitrile and 0.1% formic acid, respectively. The gradient was started as 97% buffer A and 3% buffer B for 3 min, followed by 3–25% buffer B for 120 min, 90% buffer B for 7 min, and finally 97% buffer A for 7 min. Data acquisition in the mass spectrometer was set to the positive ion mode, with a selected mass range of 300–2000 m/z . Tandem mass spectrometry was performed on peptides with +2, +3, and +4 charge states. ProQuant software v1.1 (Applied Biosystems, MDS-Sciex) was used for protein quantification and identification. Identifications were made using the *P. marinus* protein database downloaded from JGI in November 2005 (http://genome.jgi-psf.org/finished_microbes/prom4/prom4.home.html). Search parameters were set at a mass tolerance of 1.2 Da, MS/MS tolerance of 0.6 Da, one missed cleavage of trypsin, oxidation of methionine, and cysteine modification of MMTS. Peptides with above 80% confidence were used for identification and quantification, and ProGroup Viewer software v1.0.6 was used to identify proteins with >95% confidence. The expression ratios were calculated, as well as a probabilistic *p*-value measure and weighted normalized error-factor, for further quantification analysis. The entire iTRAQ experiment was conducted twice using different cultures (biological replicate) to gain statistical evidence for differential expression of proteins in changing light intensities.

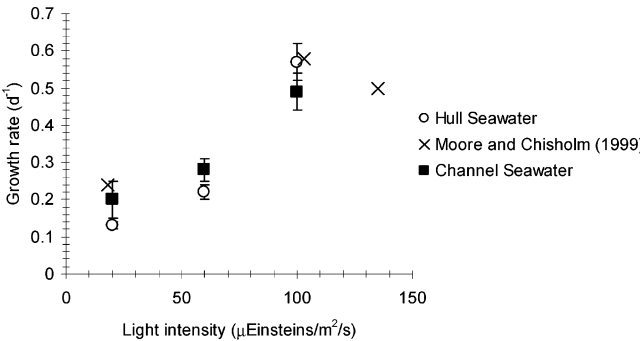


Figure 1. Growth rate as a function of light intensity. The growth rate of *P. marinus* MED4 was compared between cultures grown in Pro99 medium made with seawater from Hull, U.K., and Channel seawater (conducted in this study), and a previous study conducted by Moore and Chisholm.³¹ The results from this study are presented as the average $\pm$ standard deviation based on triplicate measurements.

Protein Data Analysis. All identified proteins were organised into functional groups using annotation based on clusters of orthologous genes, COG's (http://genome.jgi-psf.org/finished_microbes/prom4/prom4.home.html). The cellular location of these proteins was predicted using the program PSORTb v2.0.³⁵ Lipoproteins are membrane proteins and cannot be predicted using this software, and therefore, LipoP v1.0³⁶ was used, which identifies a lipoprotein by a specific signal sequence. The hydrophobic character of all the proteins identified was expressed using the GRAVY index³⁷ and was used to support the predictions of any integral membrane proteins. Finally, the abundance of each protein was estimated using the exponentially modified protein abundance index (emPAI).³⁸ EmPAI is exponentially related to PAI, which is defined as the number of observed unique peptides per protein normalized by the number of peptides obtained by *in silico* digestion. *In silico* digested peptides were generated using Proteogest software.³⁹

Results and Discussion

Growth Studies. *P. marinus* MED4 was cultured in the laboratory in three different light intensities, low (20 $\mu\text{Einstein m}^{-2} \text{ s}^{-1}$), medium (60 $\mu\text{Einstein m}^{-2} \text{ s}^{-1}$), and high (100 $\mu\text{Einstein m}^{-2} \text{ s}^{-1}$). The growth rate increased with increasing light intensity. In low light, the growth rate was $0.13 \pm 0.01 \text{ day}^{-1}$, in medium light, $0.22 \pm 0.02 \text{ day}^{-1}$, and high light, $0.57 \pm 0.05 \text{ day}^{-1}$. These growth rates are comparable to those seen in cultured isolates by Moore et al.,³¹ where the growth rates of 10 *Prochlorococcus* strains were measured by enumerating cells with a FACScan flow cytometer (Figure 1). The growth rate comparisons between both sources of seawater (Hull, U.K. and Channel), also shown in Figure 1, show that *Prochlorococcus* MED 4 grows at remarkably similar rates, although Hull seawater contains 250% more iron and higher concentrations of limiting elements (see Supporting Information for ICP-MS comparison of the different seawater).

Overview and Protein Properties. The proteins identified in this study were characterized and compared to findings in similar proteomic experiments conducted in cyanobacteria. Two iTRAQ experiments were conducted and will be referred to as iTRAQ1 and iTRAQ2. In total, 184 proteins were identified; 53 proteins were identified in both experiments, 94 unique to iTRAQ1, and 37 unique to iTRAQ2. This total corresponds to ca. 11% of total putative protein coding genes. This is a similar

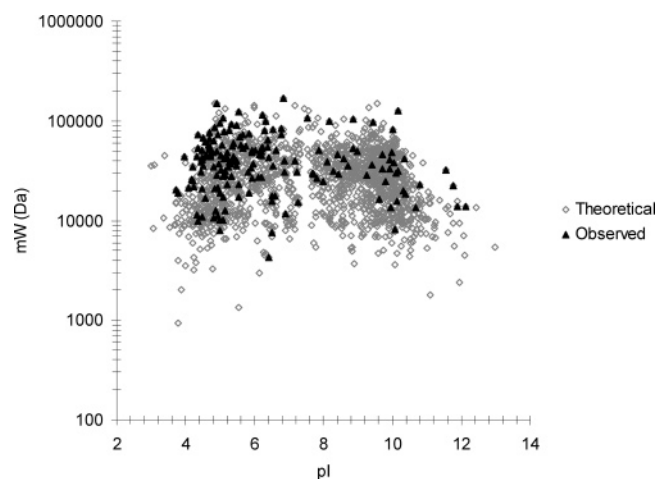
Table 2. Number and Percentage of Total Identified Proteins Grouped by Clusters of Orthologous Groups (COG)

code	COG group	identified proteins	% of total
Metabolism			
C	Energy production and conversion	16	8.2
G	CHO transport and metabolism	12	6.1
E	AA transport and metabolism	17	8.7
F	Nucleotide transport and metabolism	6	3.1
H	Coenzyme metabolism	9	4.6
I	Lipid metabolism	4	2.0
Q	Secondary metabolite biosynthesis, transport, and catabolism	2	1.0
Information Storage and Processing			
J	Translation, ribosomal structure, and biogenesis	24	12.2
K	Transcription	10	5.1
L	DNA replication, recombination, and repair	7	3.6
Cellular Processes			
D	Cell division and chromosome partitioning	5	2.6
O	PTM, protein turnover, chaperones	17	8.7
M	Cell envelope biogenesis, outer membrane	5	2.6
N	Cell motility and secretion	0	0.0
P	Inorganic ion transport and metabolism	7	3.6
T	Signal transduction mechanisms	5	2.6
R	General function prediction only	7	3.6
S	Function unknown	7	3.6
No COG		36	18.4

proteome coverage seen using a typical shotgun approach in cyanobacteria such as *Anabaena variabilis* ATCC 29413 (ca. 13%)⁴⁰ and *Synechocystis* sp. PCC6803³³ (ca. 11%). Approximately 40% of proteins were identified with 2 or more peptides. Where only 1 peptide was identified, 38% of them were identified several times (in different MS/MS experiments). Every functional category based on COG⁴¹ was represented (groups C to S), apart from group N (cell motility and secretions), which consists of only two possible proteins (Table 2). The most significant fraction of identified proteins (18.4%) could not be classified into COGs and requires further annotation. The highest represented and annotated proteins were involved in translation, ribosomal structure, and biogenesis (12.2%). A total of 33.7% of proteins was involved in metabolism, which is surprisingly similar to the 33% involved in cellular metabolism observed in *A. variabilis* ATCC 29413.⁴⁰

The proteins observed experimentally were mapped against the theoretical proteome in relation to MW and pI in Figure 2. Of the total proteins observed, 73.3% were acidic and 26.7% basic. This was an unexpected result because the theoretical proteome of *P. marinus* MED4 is 60% basic, implying that the techniques used are biased toward acidic proteins. Whether this is the result of sample preparation prior to or during iTRAQ, or the iTRAQ procedure itself, is currently not known. If we considered only the acidic proteome in this study, coverage would be 18.5% of the total protein coding genes. A total of 71.1% of identified proteins had a pI value between 4 and 7; the lowest pI was 3.72 (hypothetical protein PMM0790), and the highest pI was 12.12 (30S ribosomal protein S11, a ribosomal structure protein). This is very similar to the range seen in other cyanobacteria such as *A. variabilis* ATCC 29413⁴⁰ (3.72–12.02) and *Synechocystis* sp. PCC6803³³ (3.44–11.81). The observed MW range in *P. marinus* MED4 was 4.2 kDa (hypothetical protein PMM0465) to 168.1 kDa (glutamate synthase in amino acid transport and metabolism), comparable to *A. variabilis* ATCC 29413 (4.3–171.3 kDa).⁴⁰

PSORTb was used to predict the cellular location of identified proteins, and these were compared to the cellular locations of

**Figure 2.** A protein map of all theoretical proteins in *P. marinus* MED4 and proteins observed in both iTRAQ experiments. MW and pI values generated from JVirGel v.2.0 (<http://www.jvirgel.de/>).

proteins identified in cyanobacteria *Synechocystis* sp. PCC6803³³ and *A. variabilis* ATCC 29413⁴⁰ (Table 3). Despite differences in sample preparation (mainly, the extraction buffers implemented), the cellular distribution of identified proteins is very similar. Cytoplasmic proteins dominate with percentages ranging from 39.2% to 49.6% across the three organisms. A higher percentage of cytoplasmic membrane proteins was seen in *P. marinus* MED4 (11.8%) compared to both *A. variabilis* (6.8%) and *Synechocystis* sp. PCC6803 (2.6%). Even though 391 proteins were identified in *Synechocystis* sp. PCC6803,⁴⁰ only 10 cytoplasmic membrane proteins were identified, compared to 21 in *P. marinus* MED4. The genome of *P. marinus* is also less than half the size of the genome of *Synechocystis* sp. PCC6803 (1.66¹⁰ vs 3.6 Mbp (genomesonline.org)).

A limitation of PSORTb is the inability to detect membrane lipoproteins, and therefore, LipoP v1.0 was implemented to achieve this. Twelve lipoproteins were predicted in total, all containing signal peptidase I. PSORTb designated three of these proteins as periplasmic proteins (PMM0090, PMM0710, and PMM0970), one cytoplasmic membrane protein (PMM0226), and one outer membrane protein (PMM0708). The cellular location of the remaining seven predicted lipoproteins was unknown using PSORTb. This increased the overall membrane protein count in *P. marinus* MED4 from 28 to 35. Again, this is more than the membrane protein count predicted in *Synechocystis* sp. PCC6803^{33,40} and is potentially the result of the urea-based extraction buffer used in this study.

The hydrophobicity of proteins was calculated using the GRAVY index, and seven proteins with values above 0.3 were identified and regarded as hydrophobic (Supporting Information). This is congruent with the five hydrophobic proteins predicted using the GRAVY index in *Synechocystis* sp. PCC6803.⁴⁰ Of these seven proteins, PSORTb predicted six to be cytoplasmic membrane proteins, and the highest index scoring protein (0.74) was a 30S ribosomal protein with an unknown location.

The protein abundance in both iTRAQ experiments was measured using emPAI.³⁸ In both iTRAQ1 and iTRAQ2, all emPAI values were below 1.0, which contrasts with 84.8% in *A. variabilis* ATCC 29413 and 76.4% in *Synechocystis* sp. PCC6803.⁴⁰ These smaller values are attributed to the abated protein load used in the iTRAQ method. The two most abundant proteins were PSI protein Psae (0.36) and thioredoxin

Table 3. A Comparison between the Number and Percentage of Proteins Identified in Different Cellular Locations in *P. marinus* MED4, *A. variabilis* ATCC 29413, and *Synechocystis* sp. PCC6803

cellular location ^a	<i>P. marinus</i> MED4		<i>A. variabilis</i> ATCC 29413 ⁴⁰		<i>Synechocystis</i> sp. PCC6803 ³³	
	observed ^b	% of total ^c	observed ^b	% of total ^c	observed ^b	% of total ^c
Cytoplasmic	81	45.5	253	39.2	194	49.6
Cytoplasmic membrane	21	11.8	44	6.8	10	2.6
Extracellular	0	0.0	3	0.5	0	0.0
Outer membrane	3	1.7	6	0.9	1	0.3
Periplasmic	4	2.2	13	2.0	11	2.8
Unknown	75	42.1	327	50.6	175	44.8
Total	184		646		391	

^a Cellular location was predicted using PSORTb v.2.0 software (<http://www.psort.org/psortb/>). ^b Total number of proteins observed. ^c Percentage of total proteins observed.

peroxidase (0.35) in iTRAQ1, and carboxysome shell protein (0.67) and PsbO (0.57) in iTRAQ2. Thioredoxins are involved in energy metabolism and were also identified as highly abundant in *A. variabilis* ATCC 29413.⁴⁰ It is apparent from this analysis using emPAI values that there are less abundant proteins in *P. marinus* MED4 compared to other cyanobacteria. This can be advantageous in detecting lower-abundance proteins as, for example, there are no phycobilisome proteins in *Prochlorococcus*, and these have been singled out as a major interference in protein detection in cyanobacterial cells.⁴⁰

Technical Variation and False-Positive Rate. The variation between technical replicates was assessed by calculating the coefficient of variation (CV) for both high-light (HL) phenotypes against the low-light (LL) phenotype (reference phenotype). The log ratios were compared for these two sets of data using all proteins identified in both iTRAQ experiments. In iTRAQ1, log ratios 114:115 and 117:115 (Table 1) were compared for 147 proteins, and the mean average CV was 0.11. For iTRAQ2, log ratios of 116:114 and 117:114 (Table 1) were compared for 95 proteins and gave a higher mean average CV of 0.18. In theory, the ratio of the technical replicates (HL phenotypes) in the same experiment should be equal to 1, and therefore, the average ratio was calculated for all the proteins together, for both iTRAQ experiments. This was achieved by calculating the average of the log ratios, using the error factor as a weight, to obtain a weighted average. In iTRAQ1, the weighted average of the log ratio of the labels corresponding to both HL phenotypes was 0.01 or a ratio of 1.03. This illustrates that there is little technical variation in quantification between the same HL phenotype samples. For experiment 2, the ratio of both HL phenotypes was similar at 0.91.

When analyzing iTRAQ data, it is important to provide a measure of the extent of false-positive protein identification. Previous studies have used a double database strategy approach³³ which involves searching a single genome database and also a much larger mixed genome database, such as, a nonredundant database. Therefore, proteins identified in both searches are more likely to be correct. Subsequently, a false identification rate can be estimated by the percentage of proteins which are identified in the mixed genome database, but do not belong to the model organism. Incidentally, an iTRAQ study looking at three different organisms, *Synechocystis* sp. PCC6803, *Sulfolobus solfataricus*, and *Saccharomyces cerevisiae*,⁴² adopted the double-database search strategy, and reported a false-positive percentage very similar to the fraction of proteins that were identified in the single genome database only.⁴² However, there are disadvantages to this approach. It is not clear whether all non-overlapping proteins are definitely incorrect, and these would be regarded as false negative.

Additionally, in the larger mixed genome database search, the assignment of proteins to closely related organisms which share a high degree of homology in proteins is very possible and would lead to false positives. In this study, we used an alternative approach to acquire an indication of the false-positive identification rate using a decoy database. Mass spectrometry data files for both iTRAQ experiments were reanalyzed in ProQuant software but against the proteome of a distant model organism *Escherichia coli* K12 (retrieved from NCBI, August 2006, 4364 ORFs). This was used as the interrogator search database (decoy database) with the exact same search parameters; this method was described elsewhere.⁴³ In the first experiment, 10 peptides were identified with above 80% confidence, and these corresponded to 10 proteins with above 95% confidence in identification. These were compared to peptides identified in iTRAQ1, and two were found to be exact matches. Peptide IINEPTAAALAYGLDJ belongs to the chaperone protein dnaK2 in both *E. coli* K12 and *P. marinus* MED4, and peptide MLLPDTVGTGGDSHTR belongs to aconitate hydratase 2/B in both organisms. DnaK2 and aconitate hydratase B were identified in iTRAQ1 by 6 and 3 unique peptides, respectively, and hence were treated as true-positive identifications. The remaining 8 proteins were classified as incorrect identifications resulting in a 5.4% false-positive rate. In the second experiment, six peptides corresponding to six proteins were identified with above 95% confidence with no matching peptides to iTRAQ2, giving a 6.3% false-positive rate. Therefore, an average of 5.9% of identifications was used as a guideline of the false-positive identification rate in this study.

Differentially Regulated Proteins. In total, 15 proteins were considered as significantly up- or down-regulated. The differentially expressed proteins were selected based on the *p*-value assigned for the quantification, the magnitude of the error factor, and the relative variance (RV). In addition, the detection of more than 1 peptide, or the same peptide identified over more than one MS/MS experiment, was a prerequisite for further analyses, as suggested in the General Guideline for Proteomics Data Publication (<http://www.mcponline.org/>). The ratio of medium-light (ML) to low-light (LL) was compared between iTRAQ1 and -2 for all 53 proteins which were identified in both experiments, and because they represent biological replicates, the expected ratio was 1.0. The average RV of 1.05 was determined logarithmically, which implies there was a 5% discrepancy for relative quantification. The RV was also calculated for all high-light (HL) to LL combinations of the 53 shared proteins (114:115 in iTRAQ1 compared to 116:114 and 117:114 in iTRAQ2 and 117:115 in iTRAQ1 compared to 116:114 and 117:114 in iTRAQ2) and produced values of 2%, 7%, 2%, and 3%, respectively. However, to ensure that only proteins

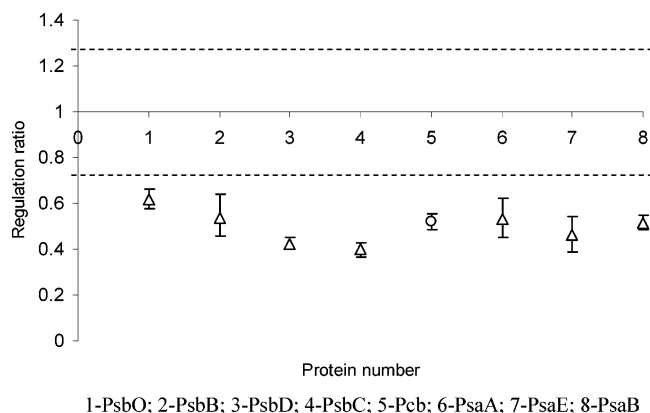


Figure 3. Differential regulation of photosystem-related proteins in high light (HL) compared to low light (LL). Proteins 1–4: Photosystem II (PsbO, PsbB, PsbD, and PsbC). Protein 5: light harvesting complex protein. Proteins 6–8: Photosystem I (PsaA, PsaE, and PsaB). Data is shown from the iTRAQ experiment in which the quantification value has the lowest *p*-value (derived in ProGroup) and, hence, is the most likely to be correct (circle, iTRAQ1; triangle, iTRAQ2). The area between the dashed lines represents the relative variance boundary (between ratios 1.25 and 0.75). The regulation ratio and standard deviation are reported in linear space.

which express a genuine differential response were selected for further analysis, only ratios of 1.25 and 0.75 (which fall clearly out of the relative variance boundaries) were deemed to be differentially regulated. The list of peptides for each differentially regulated protein was further analyzed using modified equations from ProQuant Software v1.1 (Applied Biosystems, MDS-Sciex).⁴⁴ The weighted average protein expression and weighted standard deviation were estimated using the log peptide quantification ratios, employing the percentage error factor (calculated by ProGroup Viewer) as a weight.⁴⁴ Differential regulation is shown as a ratio shown by converting out of log space.

Thylakoid Complexes. Nine proteins were identified as belonging to photosystems I (Psa) and II (Psb), including the sole light harvesting complex protein (Pcb). Eight proteins were treated as statistically significantly down-regulated in HL (Figure 3). There are several light harvesting complex genes in LL ecotypes of *Prochlorococcus*, and they are believed to be important for survival in extremely low irradiances by enabling cells to use the few available photons which reach the deeper ocean.¹⁴ However, there is only a single copy of the *pcb* gene in HL-adapted *P. marinus* MED4. Our data shows that this sole antennae protein present in MED4 is up-regulated in LL (protein 5 in Figure 3). It is therefore likely that the abundance of one specific Pcb protein in MED4, as well as the multiple light harvesting complexes in LL-adapted ecotypes, allows *Prochlorococcus* to adapt to LL. The multiple light harvesting complexes in LL-adapted *Prochlorococcus* are more sensitive to photodamage by HL conditions present at the ocean surface, demonstrated at 100 $\mu\text{einsteins m}^{-2} \text{s}^{-1}$ light intensity in the laboratory,¹³ and this may be the reason the MED4 harbors and increases the expression of only one Pcb protein. *PsbB*, *psbC*, and *psbD* encode CP47, CP43, and D2, respectively, and transcript levels for all were found to be down-regulated 50–100% in *Synechocystis* sp. PCC6803 after 15 h of exposure to HL.²³ Despite the observed down-regulation of several thylakoid located proteins, the photosystem II (PSII) reaction center core protein D1 (coupled with core protein D2), encoded by the *psbA*

gene, has been shown to increase in response to HL in MED4 in the short term, subsequently returning to normal levels.⁴⁵ Transcript levels of this gene were measured using RNA blots. However, the LL-adapted *P. marinus* SS120 strain maintains high levels of *psbA* transcripts, and it is not certain whether this difference is an important factor in the ability of MED4, unlike SS120, to acclimate to rapidly changing irradiances. However, the presence of only one copy of the D1 protein in *Prochlorococcus* strains compared to several copies found in all other examined cyanobacteria^{46,47} is thought to contribute to the reduced repair efficiency of its PSII.⁴⁸ Unfortunately, the D1 protein was not identified by either iTRAQ experiment. Despite the 0.44 and 0.55 reported down-regulation of the peptide AFINNLPFYR belonging to PSI protein PsaL, the regulation was not confirmed because only one peptide was identified in only one MS/MS experiment.

Figure 4 shows a schematic representation of thylakoid proteins with confirmed differential regulation in high light. Approximately 50% of the major proteins involved in photosynthetic activity were identified with high confidence. Overall, it appears that down-regulation of PSI and PSII proteins in HL is a response *P. marinus* adopts to reduce the susceptibility of cells to HL-induced damage.

A medium-light phenotype was included in both iTRAQ experiments to help understand the light-dependent regulation of proteins. Five of the photosynthetic proteins (PsbO, PsbC, PsbD, PsaE, and PsaL) experienced very little differential regulation from LL to ML, and then decreased in expression in HL. PsbB and the light harvesting protein (Pcb) increased in abundance in ML then decreased in HL. This change indicates that ML conditions may encourage the light harvesting capacity of cells, rather than being directly detrimental. Both major reaction center subunits in PSI, PsaA and PsaB, were down-regulated in increasing light (from LL to ML to HL) as demonstrated previously.²³ The ratio of these proteins in HL and LL in relation to ML is represented in Figure 5.

General Stress Proteins. Stress-associated proteins are vital in bacteria which suffer in adverse environmental conditions.⁴⁹ Proteins can often lose their function as polypeptides become misfolded or unfolded, and it is the job of stress-related proteins to prevent these from accumulating in the cell and, hence, prevent cellular damage. Six stress-related proteins were identified in *P. marinus* in this experiment, GroEL, GroEL2, DnaK2, HtpG, ClpC, and GroES. GroEL, HtpG, and GroES were deemed to be significantly up-regulated in HL (Figure 6). GroEL had the highest up-regulation (2.42) in iTRAQ2, whereas GroEL2 had the smallest recorded marginal up-regulation (1.36), also in iTRAQ2. Transcript levels of both these proteins were both up-regulated in *P. marinus* MED4 in a study using quantitative RT-PCR.²⁴ GroEL transcript levels rose steadily upon exposure to 200 $\mu\text{einsteins m}^{-2} \text{s}^{-1}$ and peaked after 6 h with an induction ratio of 10.11 ± 5.01 with respect to the LL culture.²⁴ GroEL2 transcript levels were comparable; however, levels continued to rise up to 12 h after HL exposure. The same study also showed transcript levels of these proteins increased in *Synechocystis* sp. PCC6803 and LL-adapted *P. marinus* MIT9313, even though they were less pronounced.²⁴ In this study, GroES was up-regulated 1.99-fold in iTRAQ2, which agrees with transcript level induction in the same microorganism.²⁴ Previous studies have shown the levels of these proteins are up-regulated during periods which immediately proceed the ‘shock’ of being exposed to HL, but the small elevated levels seen in our study are possibly a requirement of long-term

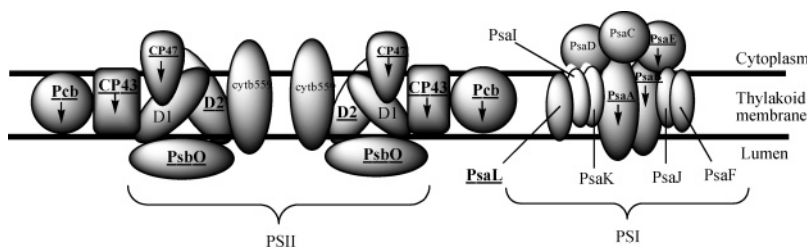


Figure 4. A schematic diagram of the major proteins located in the thylakoid membrane in MED4. Proteins in bold and underlined were identified in our study, and differential regulation in high light (HL) compared to low light (LL) is indicated by the arrow. The organization is adapted from Partensky and Garczarek.⁶⁰

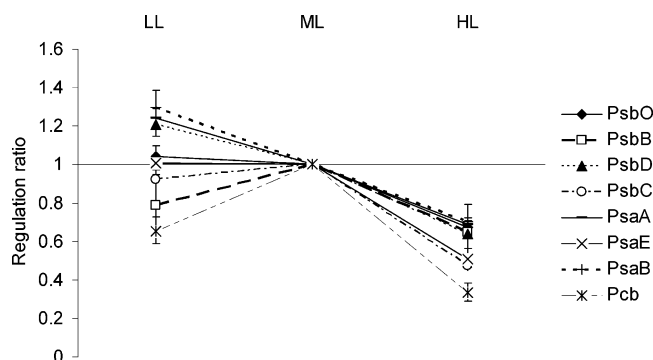


Figure 5. The relationship between medium-light (ML) phenotype and high-light (HL) and low-light (LL) phenotypes for photosystem-related proteins: Photosystem II (PsbO, PsbB, PsbD, and PsbC), Photosystem I (PsaA, PsaE, and PsaB), and light harvesting complex protein (Pcb). The value for ML is always 1 because the ratio of ML to ML is 1. Only the most probable (lowest *p*-value) quantifications are shown. The regulation ratio and standard deviation are reported in linear space.

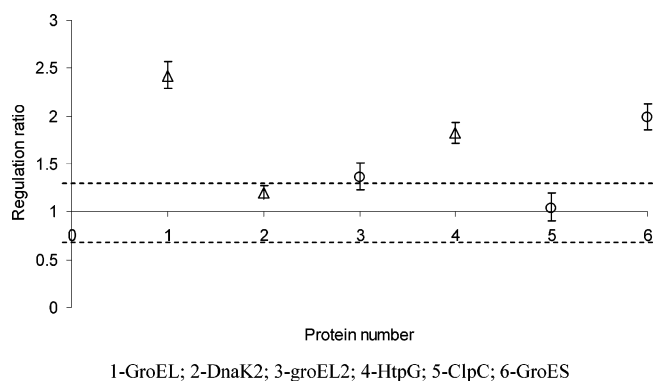


Figure 6. Regulation of stress-related proteins in high light (HL) compared to low light (LL). Proteins 1–6: GroEL, DnaK2, GroEL2, HtpG, ClpC, and GroES. Data is shown from iTRAQ experiment in which the quantification value has the lowest *p*-value (derived in ProGroup) and, hence, is the most likely to be correct (circle, iTRAQ1; triangle, iTRAQ2). The area between the dashed lines represents the relative variance boundary (between ratios 1.25 and 0.75). The regulation ratio and standard deviation are reported in linear space.

survival at high irradiances. A microarray study on HL effects on *Synechocystis* sp. PCC6803 found the transcript levels of GroEL, GroEL2, and GroES all remained high even after 15 h of exposure to HL, unlike other stress-associated genes.²³ Clearly, their presence is important to maintain cellular function, and the exact role of these chaperones in stress conditions

has been described previously in the cyanobacterium *A. variabilis*.⁵⁰

Chaperone DnaK2 is a highly conserved protein among cyanobacteria,²⁴ and it has been shown to play a significant role in adaptation to HL.^{23,24} However, the protein does not seem to be differentially expressed in either iTRAQ experiment (Figure 6 and Table 4), and hence, the role of DnaK2 in long-term adaptation to HL is questionable. Interestingly, the regulation changes were slightly more evident when comparing ML to LL with ratios 1.58 and 1.32 in iTRAQ1 and -2, respectively (Table 4).

The heat shock protein HtpG was up-regulated up to 1.82-fold in HL, as were transcripts for this protein in *P. marinus* MED4²⁴ and *Synechocystis* sp. PCC6803.^{23,24} This protein has been shown to play a very important role in cyanobacterial cell survival in thermal⁵¹ and oxidative stress.⁵² In the latter study, a *Synechococcus* sp. PCC7942 *htpG* mutant had a reduced survival rate in oxidative stress caused by high irradiance and led to photoinhibition of PSII.⁵² Our study shows that HtpG is important for the survival of *P. marinus* MED4 in extreme HL conditions on the ocean surface.

ClpC belongs to a family which contains ATP-binding proteins, which function as molecular chaperones, and regulate ATP-dependent proteolysis. They are thought to protect cells from stress by controlling aggregation and denaturation of proteins, and transcript levels have been shown to increase more than 10-fold in *P. marinus* MED4 upon exposure to HL, but not in LL ecotypes.²⁴ The constitutive level of *clpC* in *Synechococcus* also increases considerably under conditions of rapid growth, for example, exposure to HL.⁵³ It is not certain whether protein levels were regulated between LL and HL cultures in this study. It was identified in iTRAQ1, and the technical replicates produced ratios of 1.04 and 1.22, suggesting little or no role in photoadaptation (Figure 6).

Figure 7 shows the regulation ratios of stress-associated proteins in HL and LL in relation to ML. Levels of GroEL and HtpG did not change significantly in ML, and this implies that unlike HL, ML does not initiate stress in *P. marinus* MED4 cells. There was no significant relative change in GroEL2 and ClpC across the three different light intensities, increasing the likelihood that these proteins are not required in photoadaptation in this cyanobacterium.

Other Differentially Regulated Proteins. Proteins not associated with the thylakoid membrane or stress-related proteins were also found to be altered by HL. The growth rate of *P. marinus* MED4 increased as the amount of light increased, and this was reflected by an increase in the FtsZ cell division protein. This protein initiates cell division by assembling into a cytokinetic ring on the cytoplasmic membrane where division occurs.⁵⁴ This protein was up-regulated in HL in iTRAQ1 and

Table 4. Relative Regulation Ratios of Additional Important Proteins

protein name	accession no.	conditions ^a	iTRAQ1		iTRAQ2	
			regulation ratio (linear space)	regulation ratio ± SD (log space)	regulation ratio (linear space)	regulation ratio ± SD (log space)
DnaK2	PMM1704	HL/LL	1.34	0.13 ± 0.05	1.06	0.03 ± 0.05
		HL/LL	1.20	0.15 ± 0.03	1.2	0.08 ± 0.02
		ML/LL	1.58	0.20 ± 0.06	1.32	0.12 ± 0.02
ClpC	PMM1088	HL/LL	1.04	0.02 ± 0.06		
		HL/LL	1.22	0.08 ± 0.08		
FtsZ	PMM1309	HL/LL	1.68	0.22 ± 0.06	1.67	0.22 ± 0.08
		HL/LL	1.62	0.21 ± 0.10	1.89	0.28 ± 0.07
		ML/LL	1.91	0.28 ± 0.13	1.24	0.09 ± 0.01
MinD	PMM0321	HL/LL			2.03	0.31 ± 0.05
CsoS1	PMM0549	HL/LL			3.02	0.48 ± 0.10
Rubisco	PMM0550	HL/LL	0.98	−0.01 ± 0.07	1.18	0.07 ± 0.02
		HL/LL	0.89	−0.05 ± 0.06	0.99	0.00 ± 0.00
ABC transporter	PMM0710	HL/LL	0.68	−0.17 ± 0.03	0.28	−0.55 ± 0.03
		HL/LL	0.67	−0.18 ± 0.03	0.34	−0.47 ± 0.01
Phytoene desaturase	PMM0144	HL/LL	1.35	0.13 ± 0.04		
		HL/LL	1.49	0.17 ± 0.01		

^a HL, high light at 100 μ einstein $m^{-2} s^{-1}$; ML, medium light at 60 μ einstein $m^{-2} s^{-1}$; LL, low light at 20 μ einstein $m^{-2} s^{-1}$.

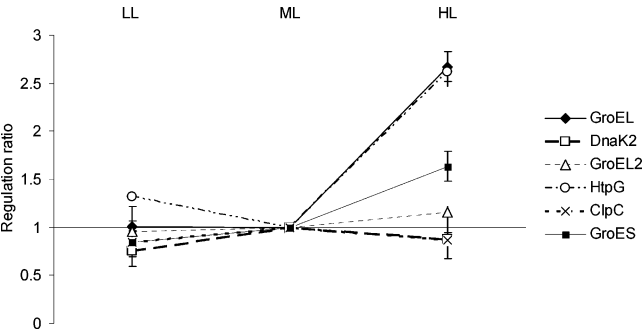


Figure 7. The relationship between ML (medium light) phenotype and HL (high light) and LL (low light) phenotypes for stress-related proteins: GroEL protein (chaperonin cpn60), molecular chaperone DnaK2 (heat shock protein), GroEL2 protein (chaperonin cpn60 2), HtpG heat shock protein, molecular chaperone ClpC, and GroES protein (chaperonin cpn10). The value for ML is always 1 because the ratio of ML to ML is 1. Only the most probable (lowest *p*-value) quantifications are shown. The regulation ratio and standard deviation are reported in linear space.

iTRAQ2, and its expression also increased in ML for both experiments (Table 4). The up-regulation of the putative septum site-determining protein MinD (PMM0321) in HL provides additional evidence for increased cell division and, hence, growth rate. This protein ensures the septum is correctly placed during cell division and was identified in iTRAQ2 and found to be 2.03-fold up-regulated.

The carboxysome is an inclusion body found in all cyanobacteria and several other microorganisms.⁵⁵ It is important for creating an elevated local CO₂ concentration⁵⁶ and contains the enzyme ribulose 1,5 biphosphate carboxylase/oxygenase (Rubisco), which converts inorganic carbon to an organic form. Carbon concentrating in the carboxysome is an energy-consuming process, and so would be expected to be down-regulated in LL. The structure and components of carboxysomes are poorly understood, and it is not clear how *Prochlorococcus* even transports inorganic carbon,⁷ but there are three carboxysome shell proteins in *P. marinus* MED4, CsoS1, CsoS2, and CsoS3. CsoS2 was identified in both iTRAQ

experiments, and CsoS1 was identified in iTRAQ2 only. The regulation of CsoS2 could not be calculated in iTRAQ1 because only one peptide in one MS experiment was detected, and the technical replicates in iTRAQ2 produced contrasting results. There was a similar case in iTRAQ2 for CsoS1; however, the *p*-value of one technical replicate was zero and gave an up-regulation of 3.02. In conclusion, the expression of these proteins remains uncertain. The expression of the enzyme Rubisco was not significantly different in the varying irradiances (Table 4), and therefore, it was concluded that carbon fixation rates did not alter. In *Synechocystis* sp. PCC6803, transcript levels for genes-encoding components of the carboxysome including Rubisco increased even after 15 h in HL.²³

ABC transporter protein (PMM0710) is part of the ABC-type phosphate transport system, and this protein was 0.68 and 0.67 down-regulated in HL in iTRAQ1, and 0.28 and 0.34 down-regulated in HL in iTRAQ2. This transporter protein is encoded by *pstS*, and this gene is known to be induced under phosphate-limiting conditions in many cyanobacteria, including *P. marinus* MED4.⁵⁷ The noninduction of PhoH-like phosphate starvation-inducible protein (PMM1284) in iTRAQ1 implied that phosphate starvation was not a factor in any of the cultures. Therefore, the mechanism by which the regulation of PMM0710 levels is related to irradiance is not presently understood, and further investigations need to be conducted to understand its role.

Carotenoids are photosynthetic pigments that play several important roles in the cyanobacterial cell. Not only do they trap light energy in the light harvesting apparatus, but they also dissipate excess light energy helping to protect the cell from radiation damage. They also provide protection from free radicals formed in the cell from a variety of stress inducers. The major carotenoids in *Prochlorococcus* are α -carotene, zeaxanthin, and a type of phycoerythrin.^{58,59} The enzyme phytoene desaturase in *Prochlorococcus* converts phytoene to ζ -carotene in the zeaxanthin biosynthetic pathway. This enzyme (PMM0144) was identified in iTRAQ1 and was slightly up-regulated in HL, 1.35- and 1.49-fold. An increase in abundance of this enzyme suggests carotenoids play a role in

adaptation to HL in *P. marinus* MED4; however, further investigations will have to be undertaken to confirm its specific function.

Conclusions

In this study, we sought a global proteomic perspective on the adaptability of the environmentally significant cyanobacterium *P. marinus* MED4 to the varying light intensities it faces in vast oceanic niches. Increasing light intensity from 20 to 100 $\mu\text{Einsteins m}^{-2} \text{ s}^{-1}$ more than quadrupled the growth rate of *Prochlorococcus* cells from 0.13 ± 0.01 to $0.57 \pm 0.05 \text{ day}^{-1}$. In total, 184 proteins, corresponding to approximately 11% of the complete theoretical proteome, were identified, and isobaric tags were used for relative quantification. Average relative variance was calculated to be 5% from biological replicates across two separate iTRAQ experiments, and the average mean CV was 0.15 for technical replicates. This meant that 15 differentially expressed proteins were identified with a high degree of confidence ($\geq \pm 25\%$ change). The initial response of *P. marinus* MED4 cells to HL is to protect photosystem-related proteins (ca. 50% were identified), including the Pcb light harvesting complex, by reducing their expression. This potentially avoids the absorption of excess light energy. Differential regulation occurred in ML, and this could be due to the role of changing stoichiometry of PSI and PSII in different light conditions, which has previously been described at the genetic level in the cyanobacterium *Synechocystis* sp. PCC 6803.²³ The increased growth rate requires elevated amounts of cell division proteins, for example, the average 1.72-fold up-regulation of the FtsZ cell division protein. The stress-related chaperone GroEL exhibited the largest up-regulation in HL (2.4), demonstrating that, despite the increased growth rate in HL, constant protection is required against the detrimental effects.

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Supporting Information Available: Tables listing the proteins and peptides identified in the iTRAQ1 and -2 experiments; ICP-MS comparison of the different seawater samples. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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